MINERALS

Dr. Humaira Aman Ali

Minerals

- Minerals are essential for the normal growth and maintenance of the body.
  - **MACROELEMENTS:** If the daily requirement is ***more than 100 mg***, they are called major elements or macrominerals or macroelements.

  - **TRACE ELEMENTS:** If the requirement of certain minerals is ***less than 100 mg/day***, they are known as microelements or microminerals or trace elements.

Important Minerals

Major elements	Trace elements
1. Calcium	1. Iron
2. Magnesium	2. Iodine
3. Phosphorus	3. Copper
4. Sodium	4. Manganese
5. Potassium	5. Zinc
6. Chloride	6. Molybdenum
7. Sulfur.	7. Selenium
	8. Fluoride.

IRON

Dr. Humaira Aman Ali

IRON

- Iron is one of the most important trace elements in the body.

- The World Health Organization (WHO) has estimated that the global prevalence of anemia to be **~24.8%**. The most common causes of anemia include nutritional deficiencies, particularly **iron deficiency**.

- Total iron content in a human varies approximately from 3 gm to 4 gm.

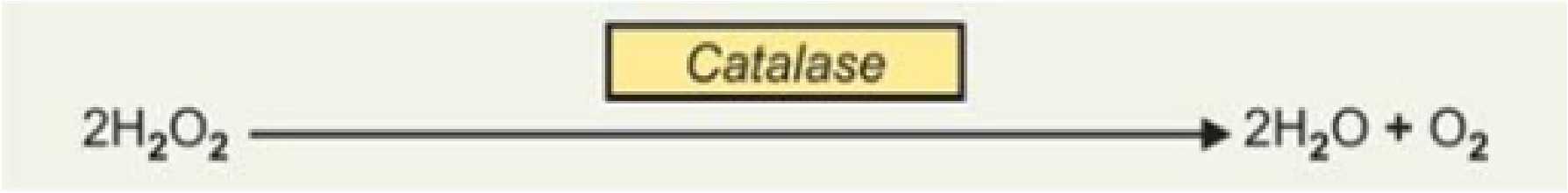
Distribution of Iron in the body

Protein/Enzyme	*Iron content (in mg)*	*% of total*
Haemoproteins		
• Haemoglobin	2500	60–70
• Myoglobin	400	5–10
• Haem enzymes:		
Catalase and peroxidase	2–3	1
Organo-iron compounds		
• Cytochromes	4–5	1
Storage iron		
• Ferritin	300–700	10–15
• Haemosiderin		
(Non-haem protein)		
Transferrin		
(Non-haem protein)	6–8	1

Haem Proteins:

• Haemoglobin and Myoglobin: These are haem proteins, which are proteins with an iron porphyrin prosthetic group attached to the protein globin. They are **most abundant** of the essential (or functional) iron compounds in the body.

• Catalases: A haem containing enzyme. It destroys hydrogen peroxide, H_2O_2, formed in the tissues and molecular O2 is evolved in the reaction.

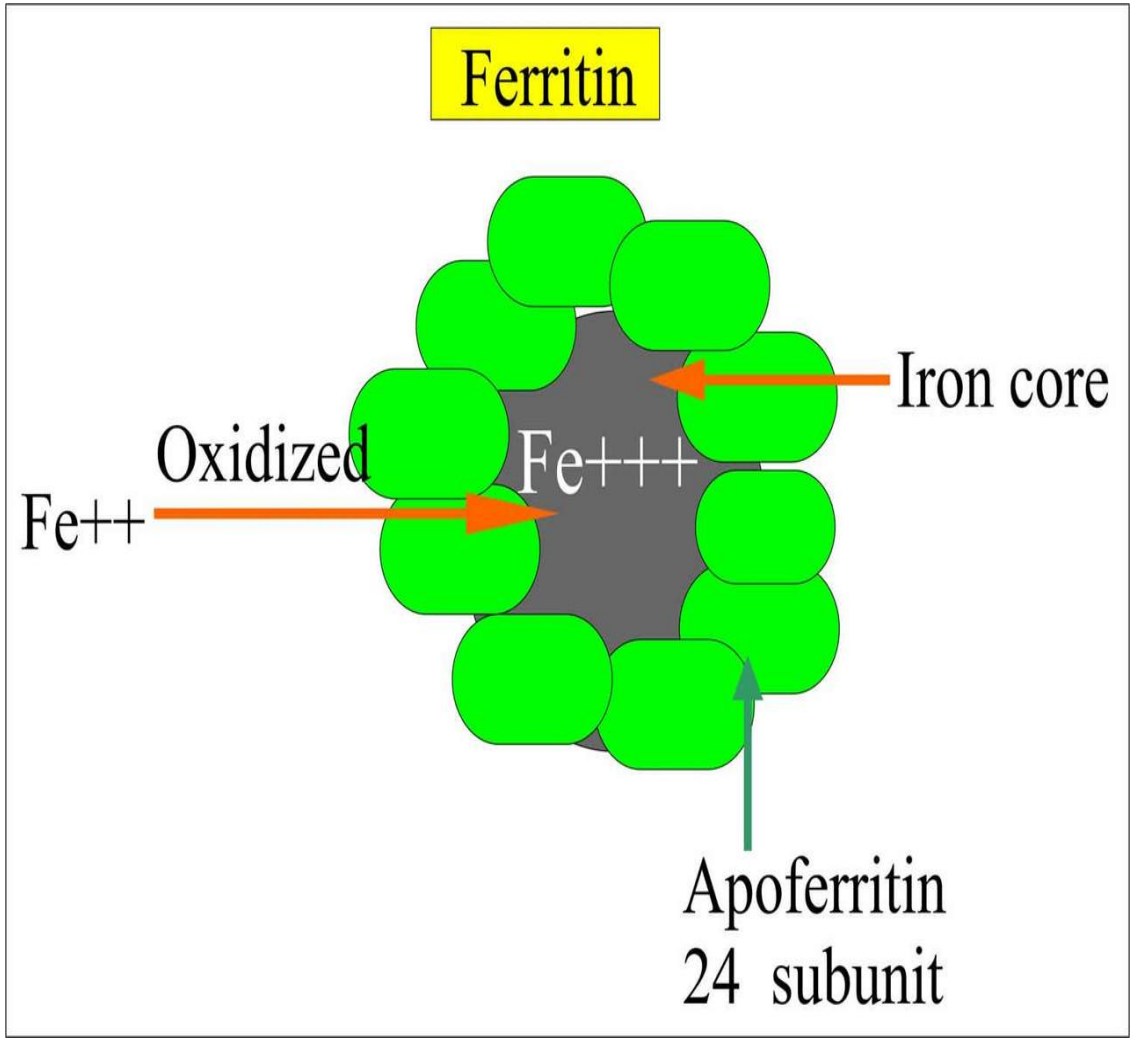


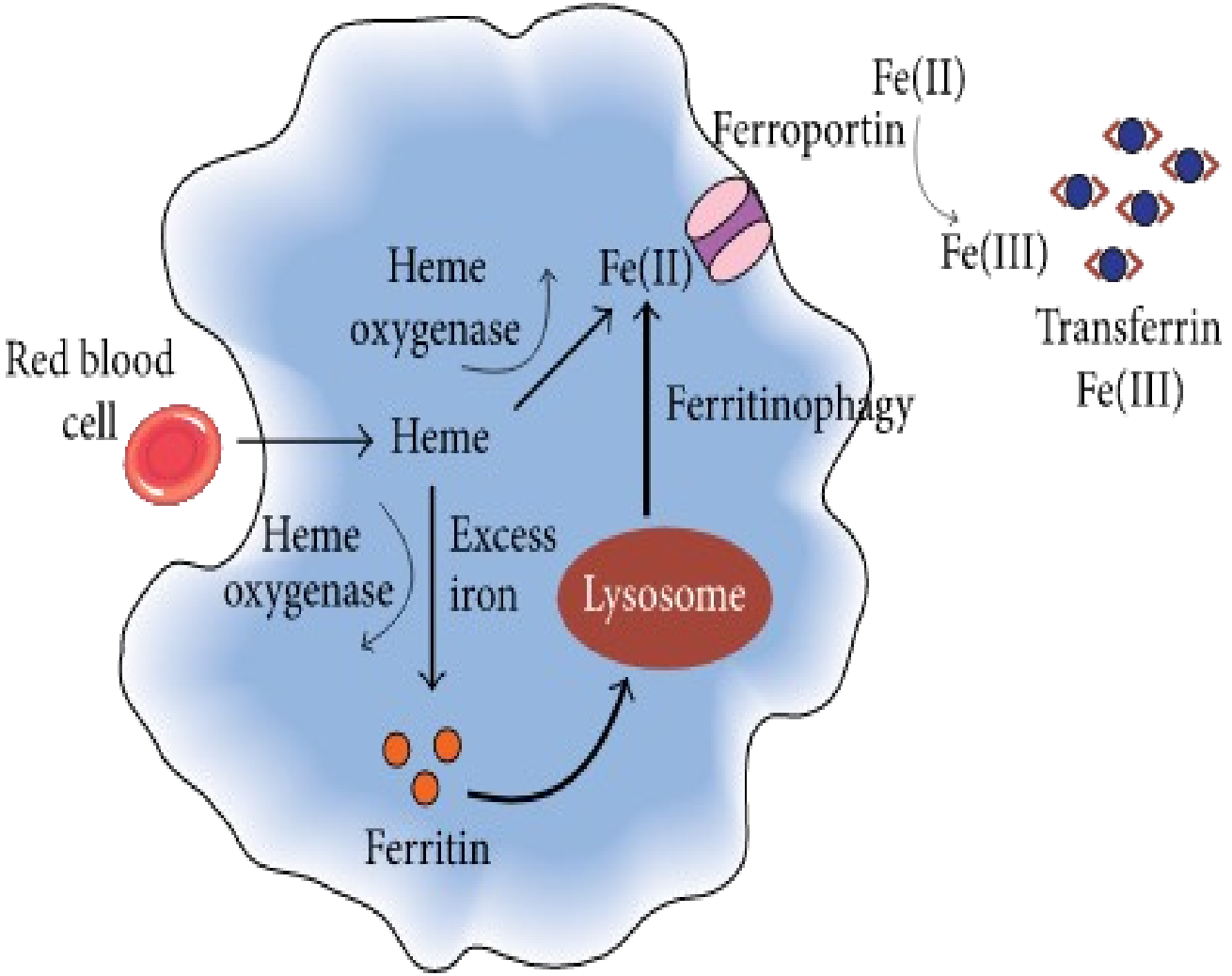
Storage Iron

- Storage iron is present in two major compounds. They are:
    1. Ferritin
    2. Haemosiderin

Ferritin:

- In the cell cytoplasm, iron combines mainly with a protein “apoferritin”, to form ferritin. Ferritin is the major protein used for intracellular storage of iron.
- Free iron is toxic and generates hydroxy radicals. Iron bound to ferritin is nontoxic.
- It is found in blood, liver, spleen, bone marrow and intestine (mucosal cells).
- Bound form of iron with ferritin is more soluble and iron is present as “ferric phosphate”.


Ferritin
Iron core
Oxidized
Fe++
Fe+++
Apoferritin
24 subunit


Fe(II)
Ferroportin
Fe(III)
Transferrin
Fe(III)
Heme
oxygenase
Fe(II)
Red blood
cell
Heme
Ferritinophagy
Heme
oxygenase
Excess
iron
Lysosome
Ferritin

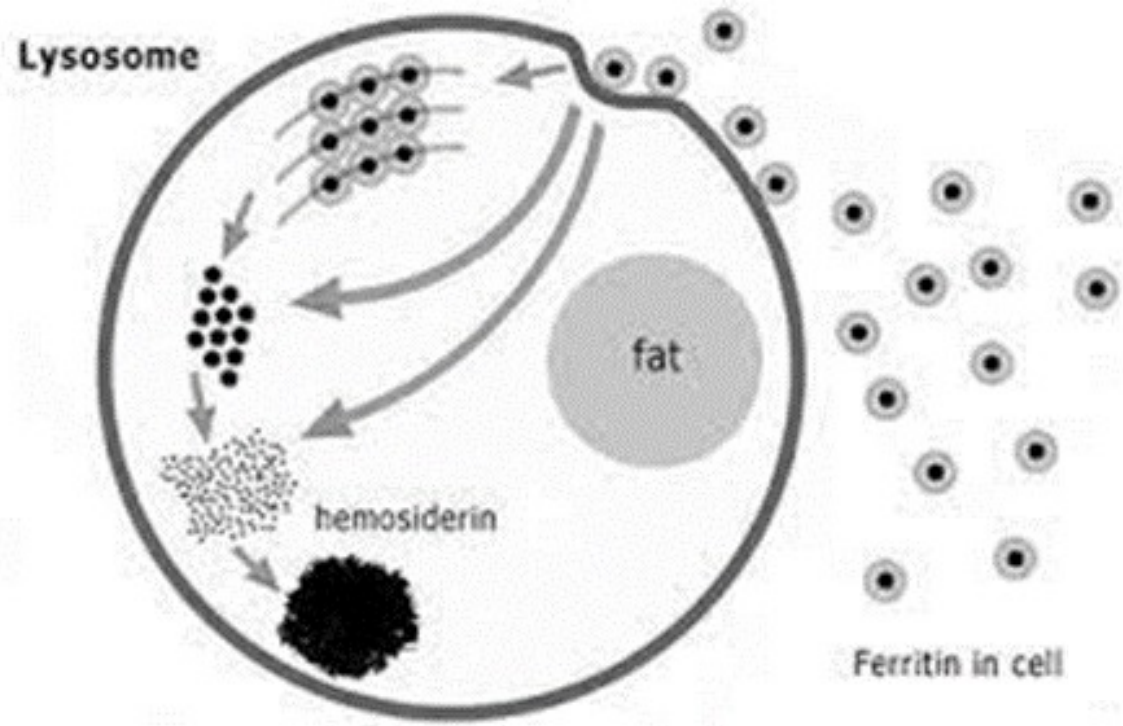
Haemosiderin:

- Hemosiderin is a yellowish-brown, iron-containing, granular pigment representing an iron storage complex that is formed by the breakdown of hemoglobin or an abnormal metabolic pathway of ferritin.

- Haemosiderin is usually seen in states of iron overload, if the storage capacity of the ferritin is exceeded.

- Haemosiderin is insoluble and mobilisation of iron is much slower from haemosiderin than ferritin.

- Hemosiderin is most commonly found in macrophages and is especially abundant in situations following hemorrhage.


Lysosome
fat
hemosiderin
Ferritin in cell

TRANSFERRIN

- Transferrin is a non-haem iron binding glycoproteins.
- It can bind with two atoms of iron in the ferric state (Fe+++).
- It exists in plasma as β1-globulin and is the true carrier of iron.
- In plasma, transferrin is saturated only to the extent of 30 per cent to 33 per cent with iron.
- Prior to binding to transferrin, Fe++ (ous) iron has to be oxidised to Fe+++(ic) form.
- Ceruloplasmin and ferroxidase II are required for this conversion. In copper deficiency, this conversion cannot occur and haemopoietic system will not be getting required amount of Fe for Hb synthesis.

Function of Transferrin

- Major function of transferrin is deliver and transport iron to all the various biological tissues between sites of absorption, utilization, and storage.
- Specific receptors are available on cells surface. Transferrin is internalised by receptor mediated endocytosis. Within the target cells, iron is released and apotransferrin is recycled to form new transferrin molecules.
- Total or transferrin iron-binding capacity (TIBC) is a test that measures the blood capacity to bind iron with transferrin.

SOURCES OF IRON

I. Exogenous: Dietary sources

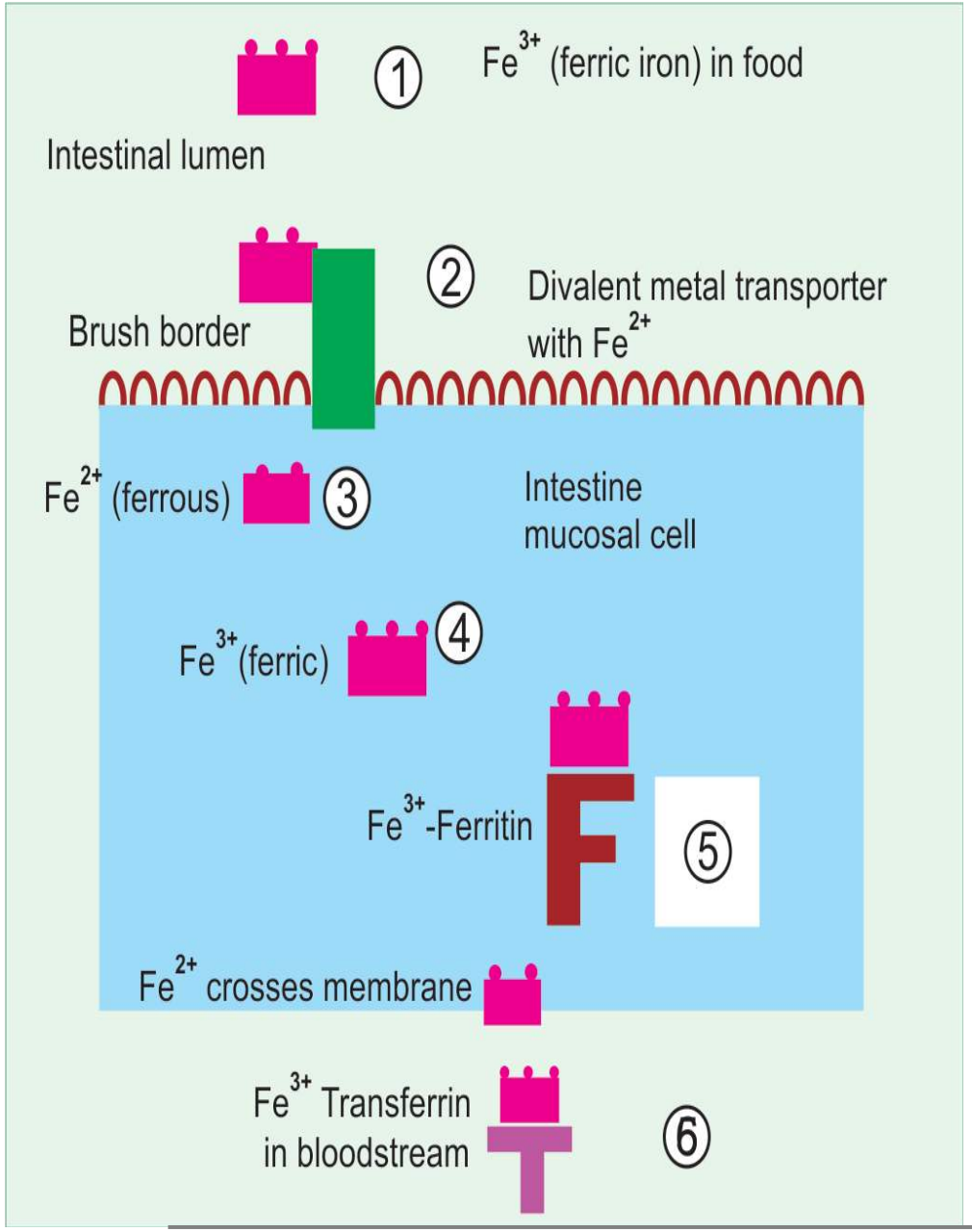
- Foods rich in iron include:
  - Animal Sources: Red meat, seafood, beef and chicken liver, are very rich sources (2.0 to 6.0 mg/100 gm). Also found in shellfish.
  - Vegetable Sources: Spinach and other leafy vegetables.
  - Cooking in iron utensils will improve the iron content of the diet.
  - Milk is a **very poor source**.

II. Endogenous: Fe is utilised from ferritin.



ABSORPTION OF IRON

- Around 10 to 20 mg of Fe is taken in the diet.
- Only about 10 % is absorbed (about 1–2 mg).
- The greatest need of iron is in infancy and pregnancy.
- Iron is absorbed by upper part of duodenum.
- Only Fe++ (ferrous) form (reduced form) is absorbed. Fe+++ (ferric) form is not absorbed.



1 = Fe+++ (ferric iron) in food.

2 = Iron is reduced to Fe++ (ferrous) state, and attaches to divalent metal transporter on the mucosal surface.

3 = Ferrous iron is internalized.

4 = Iron is oxidized to ferric.

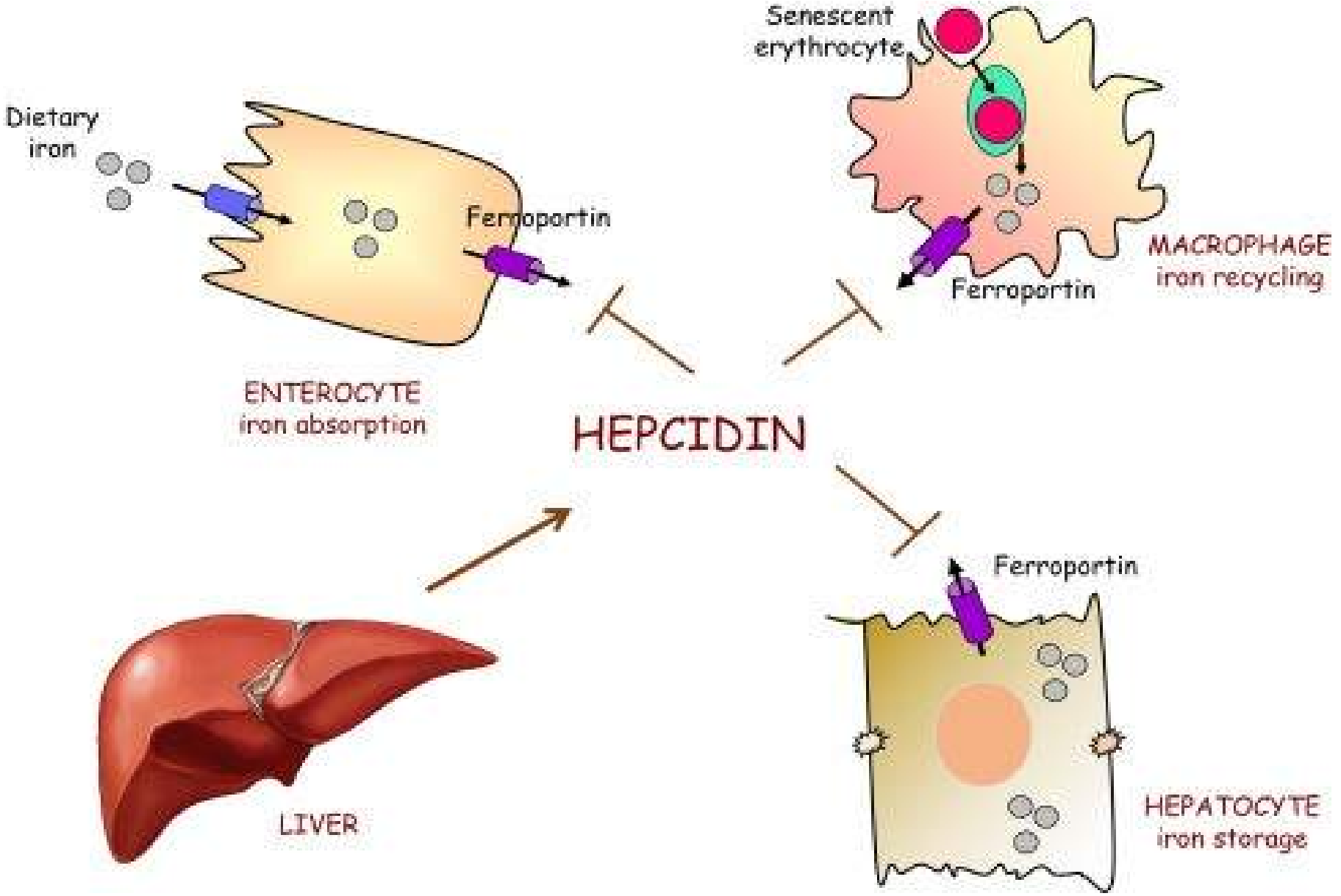
5 = Ferric iron binds with ferritin for storage.

6 = Ferric iron released, reduced to ferrous, crosses the cell membrane, re-oxidized to ferric state by ceruloplasmin.

In the blood-stream, ferric iron is bound with transferrin.

Hepcidin:

- The peptide hepcidin plays a central role in iron homeostasis.
- Synthesized in the liver , hepcidin binds to the cellular iron exporter, ferroportin, triggering its degradation.
- The consequent decrease in ferroportin produces a “mucosal block” by lowering iron absorption in the intestine and depresses recycling of the iron liberated by red blood cell turnover.
- Together, these result in a reduction in circulating iron levels.
- When plasma iron levels are high, hepatic synthesis of hepcidin increases, reducing both iron absorption and recycling.



Ferroportin is a transmembrane protein that transports iron from the inside of a cell to the outside of the cell.

FACTORS REGULATING ABSORPTION

The regulation of is done at the absorption level.

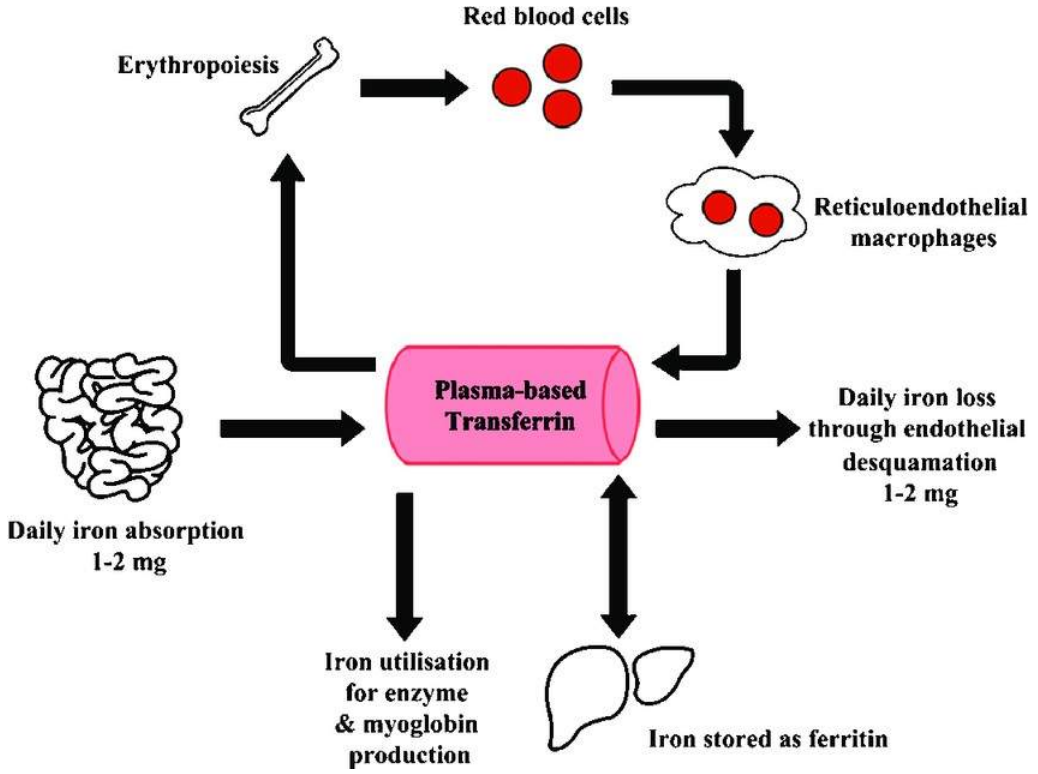
- **Factors increasing iron absorption:**
  - Ascorbic acid: Ferric ions are reduced with the help of gastric HCl and ascorbic acid.
  - Low body stores of iron: iron deficiency anemia (10 to 20%).
  - Hypoxia
  - Haem iron which comes mainly from animal products and is from Hb and myoglobin, is efficiently absorbed.
  - Non-haem iron, which is present in plants, though ingested in larger amount than haem iron, are inefficiently absorbed.

FACTORS INHIBITING IRON ABSORPTION

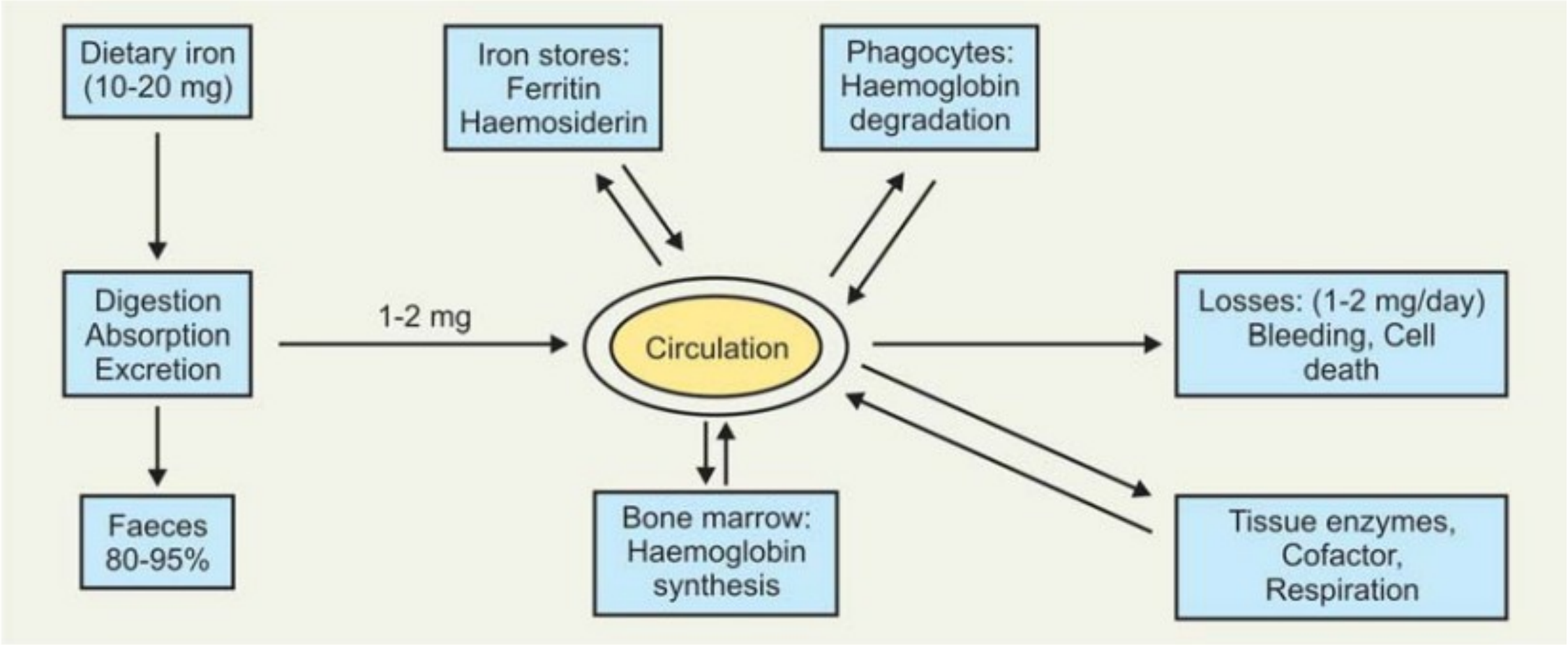
- Foods that inhibit non-haem iron absorption to some extent are:
  - Tea and coffee
  - An excess of phosphates, phytates and oxalates
- pH of duodenum: At high alkaline pH, the ingested iron is precipitated
- Antacid therapy
- Achlorhydria
- Subtotal gastrectomy and surgical removal of the upper small intestine
- Malabsorption syndrome
- Diarrheal diseases
- Chronic infections

LOSS OF IRON FROM THE BODY

- Normally, the loss of iron from the body of healthy males is limited to 0.5 mg/day through intestine.
- Menstruating women lose iron with menstrual blood (1mg/day).
- Feces contain unabsorbed iron as well as iron trapped in the intestinal cells, which are then desquamated.
- No iron is present in urine.
- Sweat also has some iron.



Summary of iron distribution and transport in humans



Iron Deficiency Anemia

Iron deficiency anaemia is manifested as hypochromic microcytic anaemia.

CAUSES:

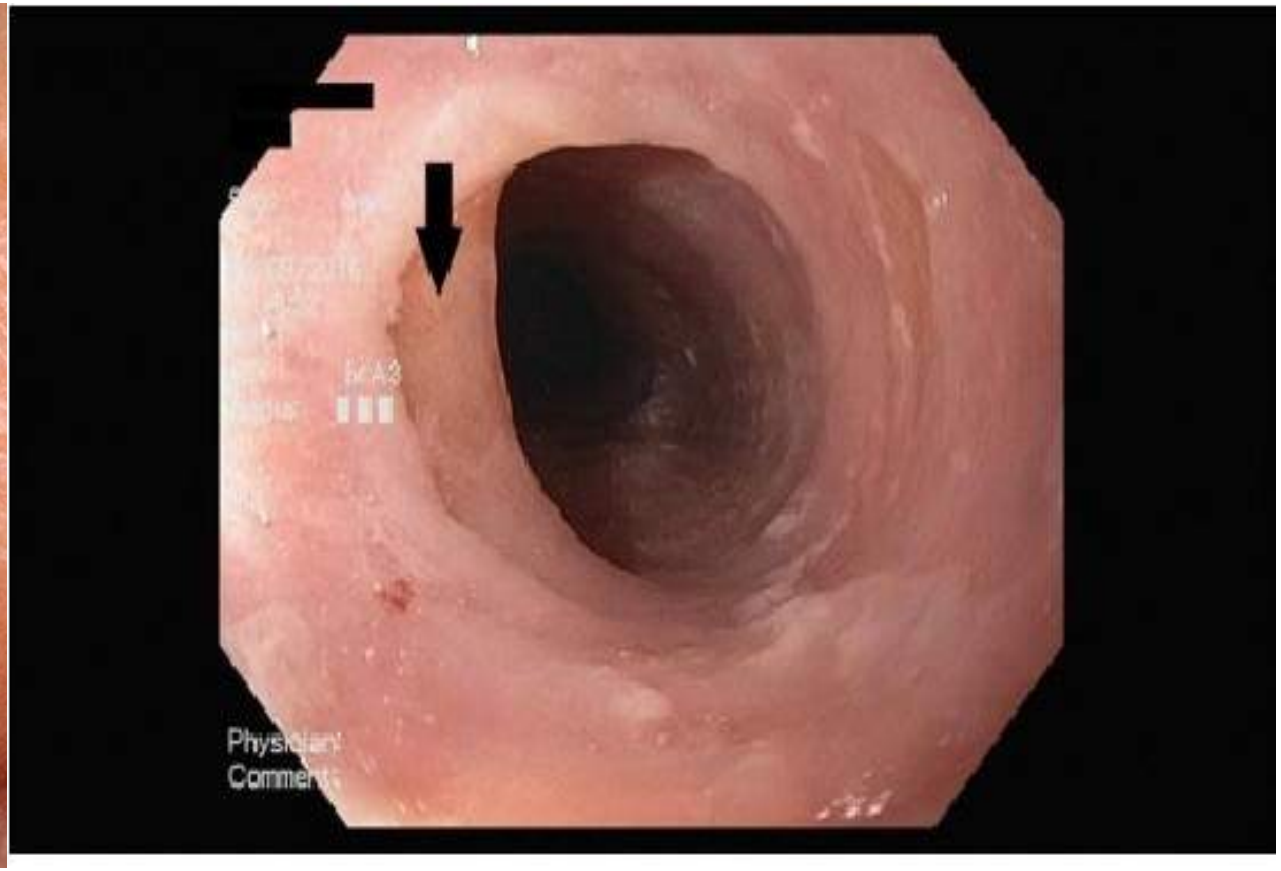
1. Chronic blood loss: Menorrhagia or bleeding gastrointestinal lesions.
2. Dietary deficiency: rare except in infants, premature newborns.
3. Increased iron requirement: in pregnancy.
4. Malabsorption: Iron is absorbed in proximal small bowel; decreased abosrption in celiac disease, inflammatory bowel disease.

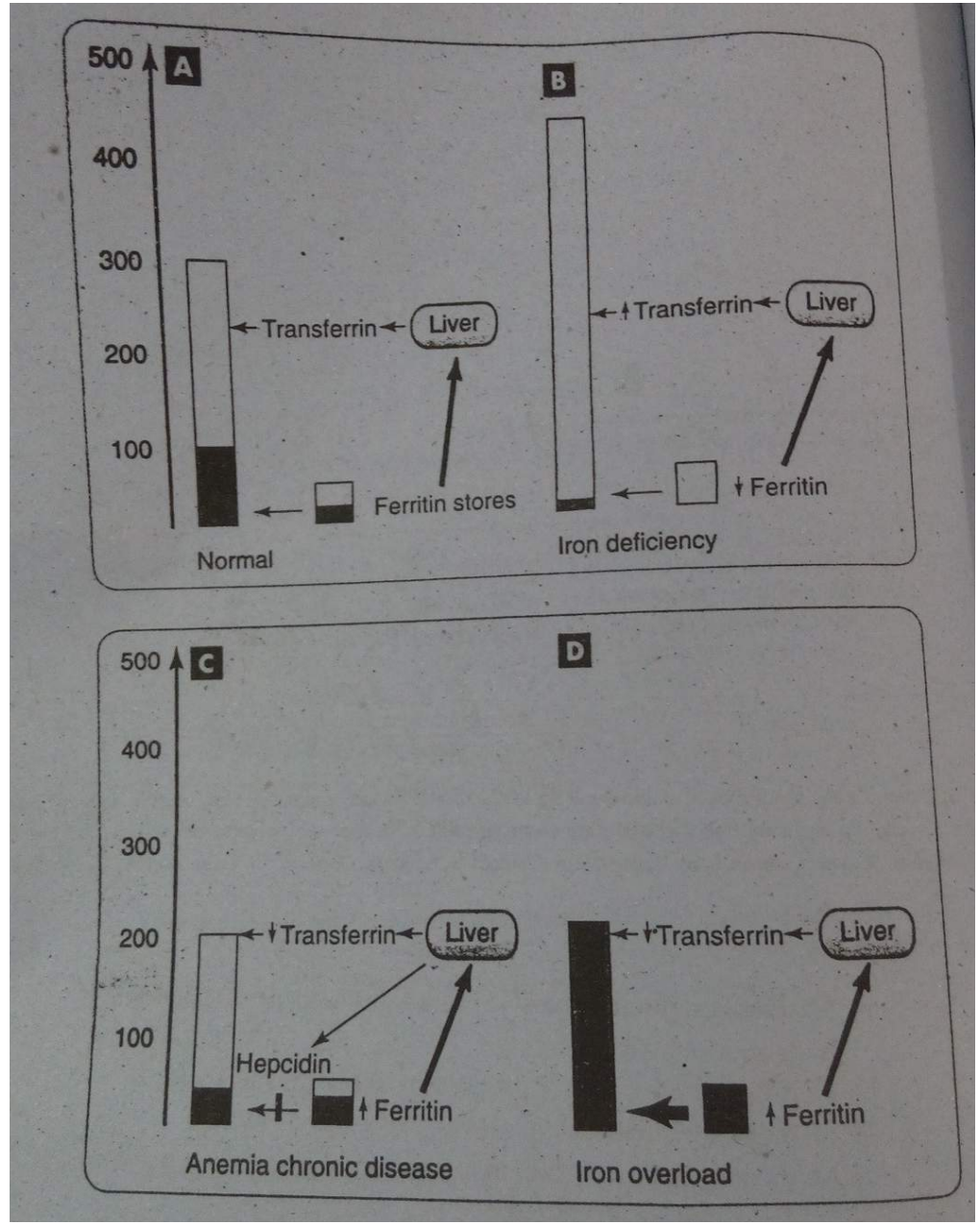
- **Clinical Manifestations:**
  - May have pallor, fatigue or dyspnea on exertion.
  - Chronic anemia: Glossitis, gastritis, koilonychia "spoon nail" Plummer Vinson Syndrome with upper esophageal web.
- **Laboratory findings:**
  - Decreased Hb, Hematocrit and RBC mass
  - Peripheral smear shows microcytic, hypochromic red cells with marked anisopoikilocytosis.
  - Decreased Serum Iron and Increased TIBC
  - Decreased body iron stores and decreased serum ferritin.

	Iron deficiency anemia	Anemia of inflammatory disease
Hematocrit	↓ to ↓ ↓ ↓	↓ to ↓ ↓
MCV	↓ to ↓ ↓ ↓	Normal to ↓
MCHC	↓	Normal
Serum iron	↓ to ↓ ↓ ↓	Normal to ↓ ↓
Serum TIBC	Normal to ↑	Normal to ↓
Serum ferritin	↓ to ↓ ↓	Normal to ↑ ↑

MCV — mean corpuscular volume, MCHC — mean corpuscular hemoglobin


ByeByeDoctor.com




500
400
300
200
100
A
Transferrin
Liver
Ferritin stores
Normal
B
Transferrin
Liver
Ferritin
Iron deficiency
C
Transferrin
Liver
Hepcidin
Ferritin
Anemia chronic disease
D
Transferrin
Liver
Ferritin
Iron overload

Iron Toxicity

- **Hemosiderosis:**
  - Iron excess is called hemosiderosis.
  - Iron overload can also be an important clinical concern.
  - Iron stores may increase due to:
    - Excessive absorption
    - Parental iron therapy
    - Repeated transfusions: Thalassemic childrens require repeated blood transfusion. If whole blood is given every time, by about 20 years of age, the patient will have hemosiderosis.

Hemochromatosis:

- **Hemochromatosis** is the abnormal accumulation of iron in parenchymal organs, leading to organ toxicity.
- Haemochromatosis can be of two types:
  - 1. Primary hereditary haemochromatosis
  - 2. Secondary haemochromatosis

- **Primary Hemosiderosis:** It is also called **hereditary hemochromatosis.**
  - Inherited disorder
  - Autosomal recessive
- In these cases, iron absorption is increased and transferrin level in serum is elevated.
- Massive accumulation of iron, mainly as ferritin and haemosiderin, in visceral organs, principally liver and skin. In the liver, hemosiderin deposit leads to death of cells and cirrhosis.
- Pancreatic cell death leads to diabetes. Deposits under the skin cause yellow-brown discoloration.
- The triad of cirrhosis with marked skin pigmentation and diabetes mellitus are referred to as **bronze diabetes.**



Dr. Humaira Aman Ali

CALCIUM (Ca)

- Calcium is an important mineral mainly found in bone and teeth.

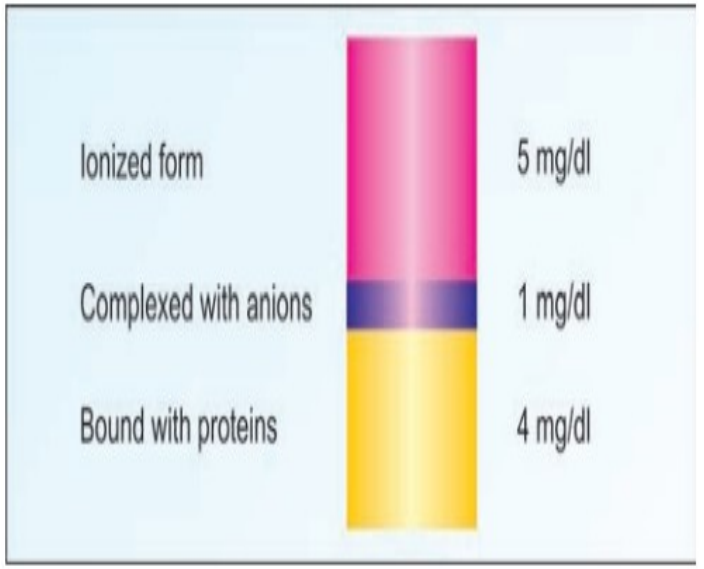
Body distribution:

- Calcium forms about 1% to 1.5% of an average adult's body weight.
- 99% of which is seen in bones and teeth as carbonate or phosphate of calcium.
- 1% of total body Ca occurs in extracellular fluid and soft tissues.

Different forms of calcium in PLASMA

- The normal level of plasma calcium is 9-11 mg/dl.
- The calcium in plasma is of 2 forms:
    - Non diffusible or Plasma protein bound calcium. Albumin is the major protein with which calcium is bound.
    - Diffusible:

1. Ionised calcium(Ca2+): 40% of total calcium is in ionised form and is physiologically active form.
2. Complexed calcium, it is probably complexed with plasma anions such as citrate and phosphate.



Dietary sources of Calcium:

- Milk and Yoghurt is a good source for calcium
- Cheese
- Egg-yolk
- Beans
- Lentils
- Nuts. Of all nuts, almonds are among the highest in calcium. Figs
- Fish and vegetables are medium sources for calcium
- Cereals (wheat, rice) contain only small amount of calcium.



Contents of Daily Calcium Requirements by Age and Sex

Age and Gender	Adequate Daily Intake of Calcium (mg)
Infants, male or female, 0–6 months	210
Infants, male or female, 7–12 months	270
Children, male or female, 1–3 years	500
Children, male or female, 4–8 years	800
Males and females, 9–18 years	1300
Males and females,* 19–50 years	1000
Males and females,† 51 years and older	1200
Pregnant or lactating female, 14–18 years	1300
Pregnant or lactating female, 19–50 years	1000

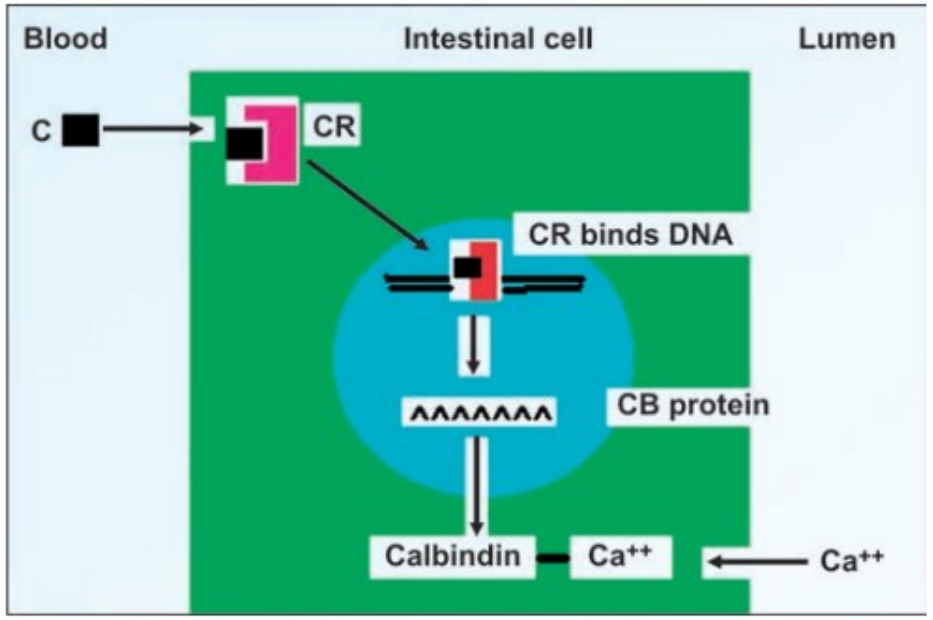
https://cdn.citl.illinois.edu/courses/KIN122/week6/chapter9/web_data/file10contents.htm

Absorption

- Calcium is absorbed mainly from the duodenum and first half of jejunum.

MECHANISM:

- Two mechanisms have been proposed for absorption of calcium by gut mucosa.
  - 1. Simple diffusion
  - 2. Calcium is absorbed against a concentration gradient, an “active” transport process requires energy and Ca++ pump.
  - Both the processes require 1, 25-di hydroxy-D3 (calcitriol).



Calcitriol increases calcium absorption.
C = calcitriol; CR = calcitriol receptor complex; CB calbindin

FACTORS AFFECTING ABSORPTION

- **Factors causing increased absorption:**
  - ✓ Vitamin D: Calcitriol induces the synthesis of the carrier protein (Calbindin) in the intestinal epithelial cells, and so facilitates the absorption of calcium.
  - ✓ Parathyroid hormone: It increases calcium transport from the intestinal cells.
  - ✓ An acidic pH favours calcium absorption because the Ca-salts, particularly PO4 and carbonates are quite soluble in acid solutions.
  - ✓ Amino acids: Lysine and arginine increase calcium absorption

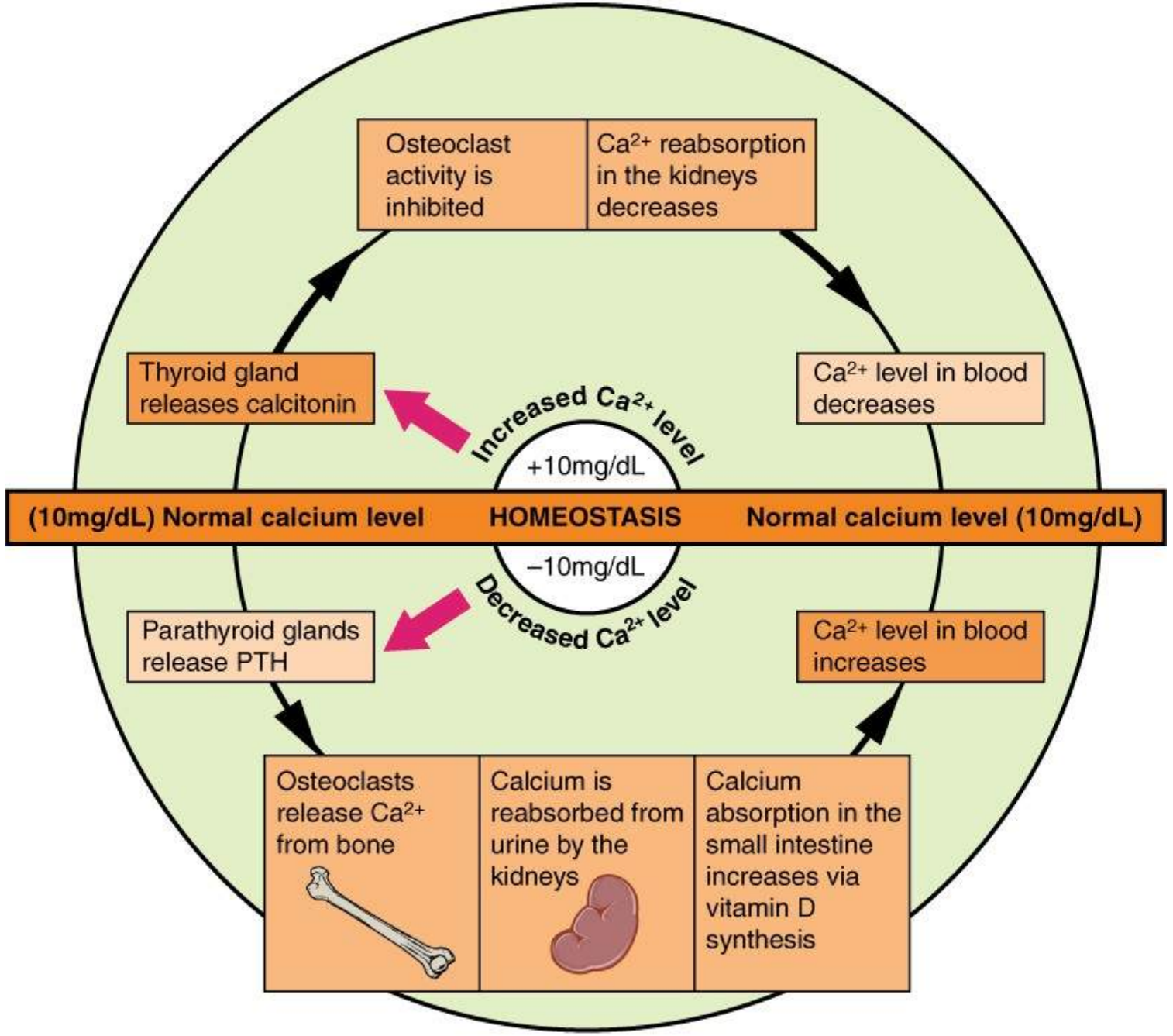
- **Factors causing decreased absorption:**

  - ✓ Phytic acid: found in plant seeds, prevents the absorption of iron, zinc, and calcium and may promote mineral deficiencies.
  - ✓ Malabsorption syndromes: Fatty acid is not absorbed, causing formation of insoluble calcium salt of fatty acid.
  - ✓ Oxalates and phosphates

FUNCTIONS

- Calcification of Bones and Teeth.
- Calcium plays a role in blood coagulation: Calcium is known as factor IV in blood coagulation cascade.
- Nerve conduction: Calcium is necessary for transmission of nerve impulses from presynaptic to postsynaptic region .
- Calcium mediates excitation and contraction of muscle fibers.
- Myocardium: Ca++ prolongs systole. In hypercalcemia, cardiac arrest is seen in systole. This fact should be kept in mind when calcium is administered intravenously. It should be given very slowly.
- Secretion of hormones: Calcium mediates secretion of insulin, parathyroid hormone, calcitonin, vasopressin, etc. from the cells.
- Activation of enzymes.

REGULATION

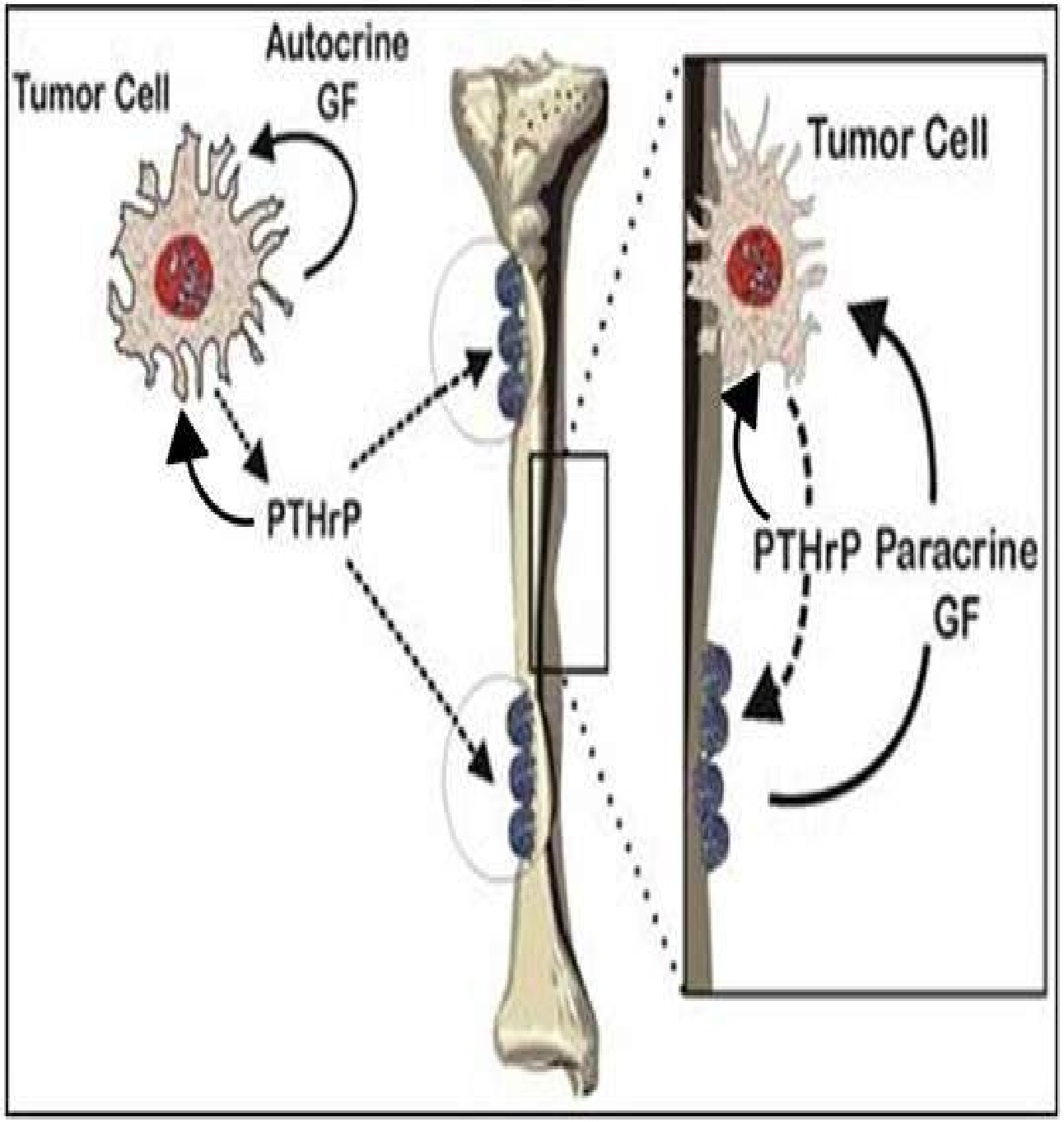

Osteoclast activity is inhibited
Ca2+ reabsorption in the kidneys decreases
Thyroid gland releases calcitonin
Increased Ca2+ level
+10mg/dL
Ca2+ level in blood decreases
(10mg/dL) Normal calcium level
HOMEOSTASIS
Normal calcium level (10mg/dL)
–10mg/dL
Decreased Ca2+ level
Parathyroid glands release PTH
Ca2+ level in blood increases
Osteoclasts release Ca2+ from bone
Calcium is reabsorbed from urine by the kidneys
Calcium absorption in the small intestine increases via vitamin D synthesis

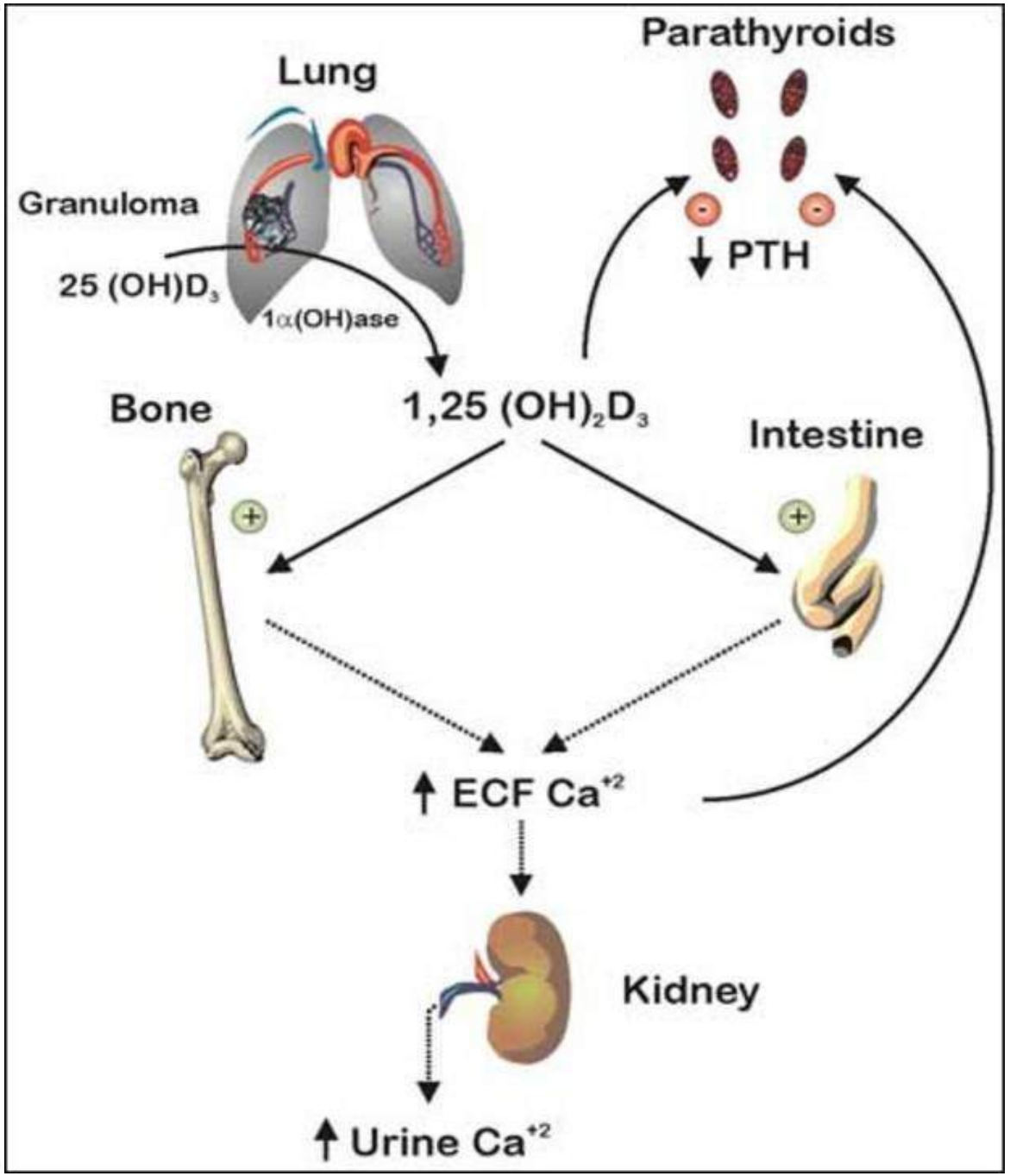
Hypercalcaemia

- When the serum calcium level exceeds **11.0 mg/dl,** it is called as hypercalcaemia.

Causes:

1. **Primary hyperparathyroidism:** Most common cause
    - Familial
    - Tumours – Solitary adenoma, Multiple adenomas, Parathyroid carcinoma
2. **"Ectopic" hyperparathyroidism:** Multiple endocrine neoplasia type I, II (MEN I) & (MEN II)
3. **Malignancy:**
    - No direct skeletal involvement:
        – PTH related protein (PTHrP)
    - Metastatic carcinoma of bone – Direct erosion of bone by tumour
4. **Haematological malignancies:** Multiple myeloma


Autocrine
GF
Tumor Cell
PTHrP
Tumor Cell
PTHrP Paracrine
GF


Lung
Parathyroids
Granuloma
25 (OH)D3
1α(OH)ase
PTH
Bone
1,25 (OH)2D3
Intestine
ECF Ca+2
Kidney
Urine Ca+2

cont..

5. **Other Endocrine Causes:** Hyperthyroidism (Thyrotoxicosis), Acute adrenal insufficiency (Addison's disease)

6. **Granulomatous Diseases:** Tuberculosis and sarcoidosis etc..

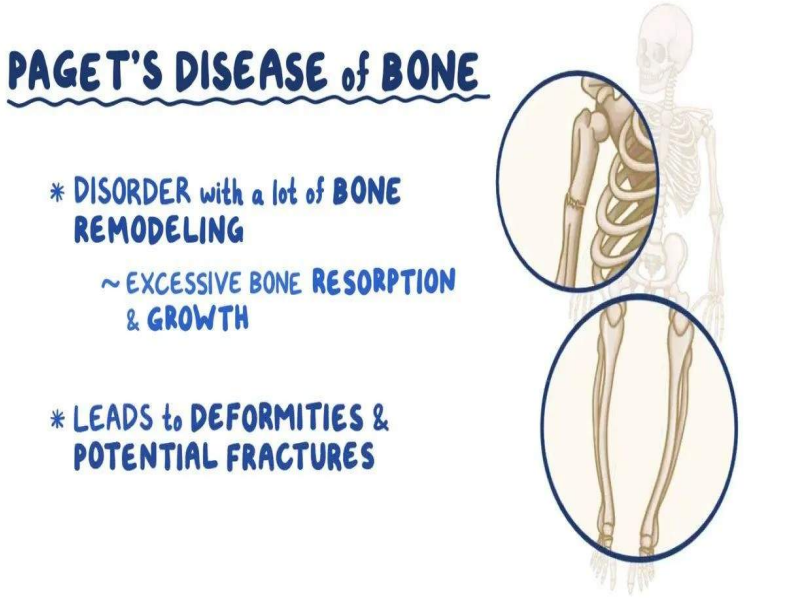
7. **Overdosage of Vitamins:** Vitamin A intoxication, and Hypervitaminosis D

8. **Drug-induced Hypercalcaemia (Iatrogenic):** Thiazide diuretics (can increase renal tubular reabsorption of calcium)

cont..

9. Miscellaneous Other Causes:

- Idiopathic hypercalcemia of infancy (William syndrome)
- Paget's disease of bone: a chronic disease characterized by episodic accelerated bone resorption and growth of abnormal replacement bone, causing bone pain and deformation
- Prolonged immobilization because of rapid bone turnover
- Milk-alkali syndrome: caused by the ingestion of large amounts of calcium supplements.



Symptoms of Hypercalcemia

1. Anorexia, nausea, vomiting
2. Polyuria and polydipsia (reduced concentrating ability)
3. Confusion, depression, psychosis and coma which can be fatal
4. Osteoporosis and pathological fracture
5. Renal stones
6. Ectopic calcification and pancreatitis
7. Abnormal heart rhythm (arrhythmia) with shortening of the QT interval.

Hypocalcaemia

- Hypocalcaemia is said to exist when serum calcium is less than 8.5 mg/dl.

Causes:

1. **Hypoalbuminemia:**
   - Reduction in serum albumin: Malnutrition, malabsorption states, nephrotic syndrome, chronic liver disease and liver failure, protein losing enteropathy

2. **Hypoparathyroidism:**
   - Surgical-induced partial or complete hypoparathyroidism (90% cases)
   - Idiopathic—may be autoimmune (10% cases)

3. **Renal Diseases and Renal Failure:**
   - Acute tubular necrosis

cont..

4. **Deficiency of Vitamin D:**
    - Decreased exposure to sunlight
    - Malabsorption
    - Dietary deficiency
    - Hepatic diseases
    - Decreased renal synthesis of calcitriol
5. **Increased Calcitonin:**
    - Medullary carcinoma of thyroid
6. **Increase in Phosphorus Level**
    - Renal failure
    - Phosphate infusion

Symptoms of Hypocalcemia

- Muscle cramps
- Paresthesia, especially in fingers
- Neuromuscular irritability, muscle twitchings
- Tetany
- Seizures
- Bradycardia
- Prolonged QT interval

Hypocalcemia causes increased neuromuscular excitability by decreasing the threshold needed for the activation of neurons. As a result, neurons become unstable and fire spontaneous action

Tetany

- Tetany is the occurrence of intermittent spasms, or involuntary contractions, of muscles, particularly in the upper and lower li the larynx.



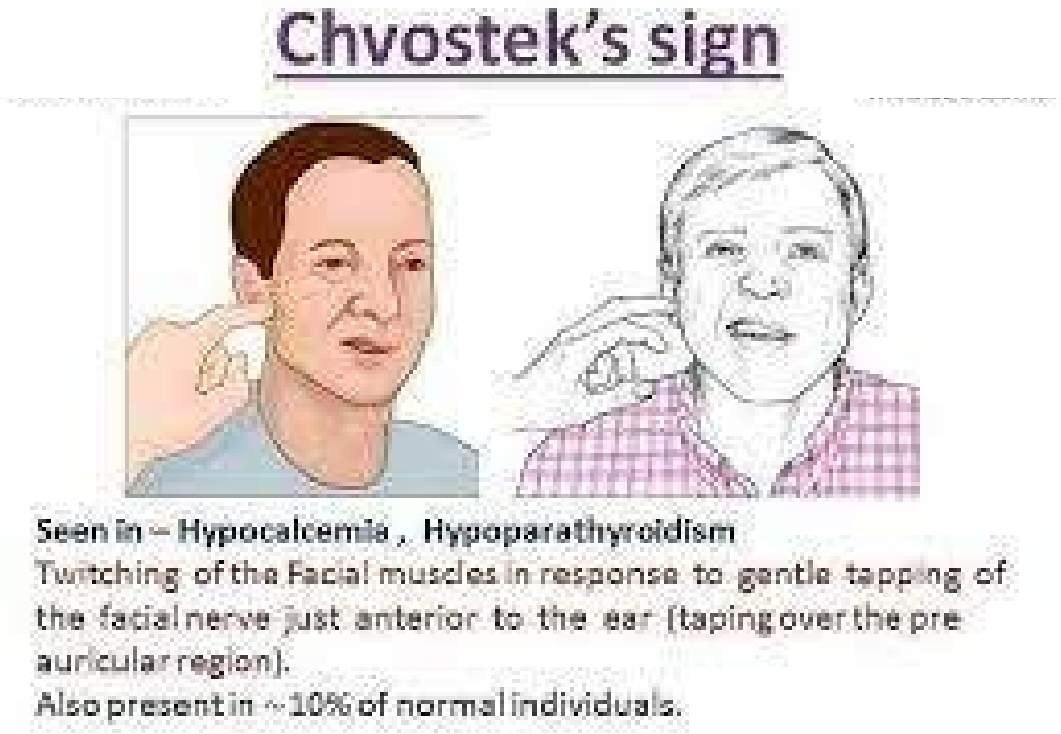
❖**Manifest tetany:** spontaneous clinical manifestations

- Classic signs of peripheral hyperexcitability of motor nerve:
  - spasm of the muscles of the wrists and ankles (carpopedal spasm) and of the vocal cords (laryngospasm)
- Sensory manifestations: paresthesia, numbness, and tingling of the hands and feet
- Motor excitability of the central nervous system:
  - brief, recurrent convulsions; usually generalized but may be localized to one side of the body

- **Latent tetany:** clinical signs not evident but can be elicited after stimulation.

- Chvostek's sign: (described by Frantisek Chvostek physician in 1876)
  - Tapping over facial nerve causes facial contraction will be positive

- Trousseu's sign: (French physician Armand Trousseau described it in 1861)
  - Trousseau’s sign for latent tetany is most commonly positive in the setting of hypocalcemia.
  - The sign is observable as a carpopedal spasm induced by ischemia secondary to the inflation of a sphygmomanometer cuff, commonly on an individual’s arm, to 20 mmHg over their systolic blood pressure for 3 minutes.
  - The carpopedal spasm is visualized as flexion of the wrist, thumb, and metacarpophalangeal joints with hyperextension of the fingers.





PHOSPHORUS

- Phosphate plays a crucial role in cell structure, signaling, and metabolism.
- Phosphate is found both as mineral phosphate(inorganic phosphate) and organic phosphate (phosphoric esters).

- **Food Sources:** Foods rich in phosphorus content are cheese, milk, nuts, organ meats, egg.

- **Body Distribution:**
  - Total body phosphate is about 500-700 g.
  - 85 % is found in bones and teeth
  - 14% in soft tissues
  - 1% is found in ECF, less than 1% circulating in serum
  - Phosphate is mainly an intracellular ion and is seen in all cells.

- **Daily requirement:** About 1 g of phosphate is required to be taken in the diet daily.

Absorption:

- 90 per cent of daily dietary phosphate is absorbed.
- The absorption is stimulated by both PTH and Vit. D3.
- Phosphorus holds an inverse relationship with calcium. An excess of serum calcium or phosphate results in the excretion of the other by the kidney.
- PTH increases calcium and phosphate release from the bone and decreases loss of calcium and increases loss of phosphate in the urine.

Functions of Phosphate Ions

1. Formation of bone and teeth.
2. Production of high energy phosphate compounds such as ATP, CTP, GTP, creatine phosphate, etc.
3. Synthesis of nucleoside co-enzymes such as NAD and NADP.
4. DNA and RNA synthesis, where phosphodiester linkages form the backbone of the structure.
5. Formation of phosphate esters such as glucose-6-phosphate, phospholipids.
6. Formation of phosphoproteins, e.g. casein.
7. Activation of enzymes by phosphorylation.
8. Phosphate buffer system in blood.

Causes of Hyperphosphatemia

1. **Increased absorption of phosphate**
   - Excess vitamin D
   - Phosphate infusion
2. **Increased cell lysis**
   - Chemotherapy for cancer
   - Bone secondaries
   - Rhabdomyolysis
3. **Decreased excretion of phosphorus**
   - Renal impairment
   - Hypoparathyroidism
4. **Hypocalcemia**
5. **Massive blood transfusions**
6. **Thyrotoxicosis**
7. **Drugs**

Causes of Hypophosphatemia

Decreased absorption of phosphate
- Malnutrition
- Malabsorption
- Chronic diarrhea
- Vitamin D deficiency

Intracellular shift
- Insulin therapy, glucose phosphorylation
- Respiratory alkalosis

Increased urinary excretion of phosphate
- Hyperparathyroidism

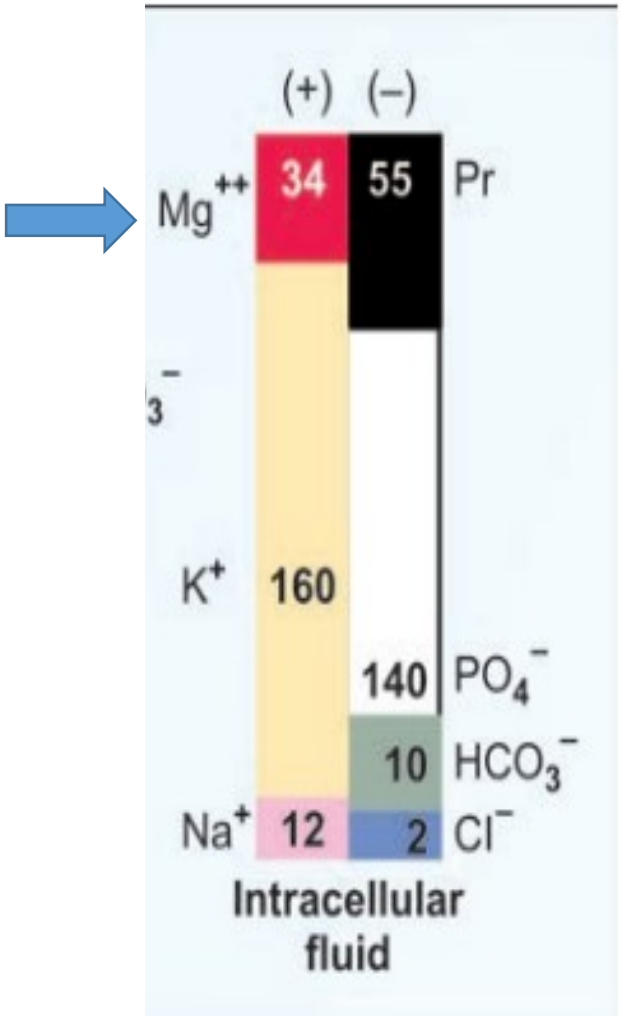
Hypercalcemia

Chronic alcoholism

Drugs
- Antacids, Diuretics, Salicylate intoxication

MAGNESIUM (Mg++)

- Magnesium is the fourth most abundant cation in the body and second most prevalent intracellular cation.
- Distribution: Approximately 2/3 occurs in bones, 1 per cent in EC fluid and remainder in soft tissues.
- Requirement:
  - The requirement is about 400 mg/day for men and 300 mg/day for women. Doses above 600 mg may cause diarrhea.
  - More is required during lactation.
  - Major sources are cereals, beans, leafy vegetables and fish.



Cont..

- About 70% of magnesium exists in free state and remaining 30% is protein-bound (25% to albumin and 5% to globulin) in plasma.

- Serum must be separated from the clot as soon as possible or the level of magnesium will increase because of its elution from the red blood cells.

- Hemolyzed samples are unacceptable for analysis.

- Excretion:
  - Magnesium is lost from the body in faeces, sweat and urine.
  - 60 to 80 per cent of orally taken Mg is lost in faeces.

Functions of Magnesium

1. Mg++ is the (cofactor) activator of many enzymes requiring ATP.
   - Alkaline phosphatase, hexokinase, fructokinase, phosphofructokinase, adenyl cyclase, cAMP dependent kinases, etc. need magnesium.

2. Neuromuscular irritability: Mg exerts an effect on neuromuscular irritability similar to that of Ca++, high levels depress nerve conduction and low levels may produce tetany (hypomagnesaemic tetany).

3. As Constituent of Bones and Teeth: About 70 per cent of body magnesium is present as in bones, dental enamel and dentin.

Hypomagnesemia

- It is commonly seen in hospital patients
- Causes of Hypomagnesemia:
  - Increased urinary loss (Tubular necrosis)
  - Hyperaldosteronism, volume expansion
  - Familial hypomagnesemia
  - Increased intestinal loss: diarrhea, laxatives, ulcerative colitis, nasogastric suction, vomiting
  - Malabsorption
  - Protein calorie malnutrition
  - Hypoparathyroidism
  - Drugs: Thiazide diuretics, Aminoglycosides, Amphotericin, Cyclosporin and Alcohol.

Hypermagnesemia

- It is uncommon and always due to excessive intake either orally (antacids), rectally (enema) or parenterally.

- Magnesium intoxication causes depression of neuromuscular system, causing lethargy, hypotension, respiratory depression, bradycardia and weak tendon reflexes.

- In severe conditions, acute rhabdomyolysis results.

Causes of Hypermagnesemia

- Excess intake orally or parenterally
- Renal failure
- Hyperparathyroidism
- Dehydration
- Drugs: Antacids and Calcitriol

SULFUR

- Sulphur is an essential element.
- The sulphur is made available to the body by the proteins containing methionine, cystine or cysteine. These amino acids contain sulphur.
- Also available from S-containing B vitamins, Thiamine (TPP), coenzyme A, Lipoic acid and biotin.
- Certain sulpholipids and glycoproteins (mucoitin and chondroitin sulphuric acid) also provide Sulphur.
- Sulphur as free element cannot be utilized.

DIETARY SOURCES

- Sulphur is available in:
  - Meat
  - Fish
  - Legumes
  - Egg
  - Liver
  - Cereals
  - Adequate protein in diet fulfills sulphur requirement

Functions of Sulfur

1. Sulfur containing amino acids are important constituents of body proteins. The disulfide bridges keep polypeptide units together, e.g. insulin, immunoglobulins.

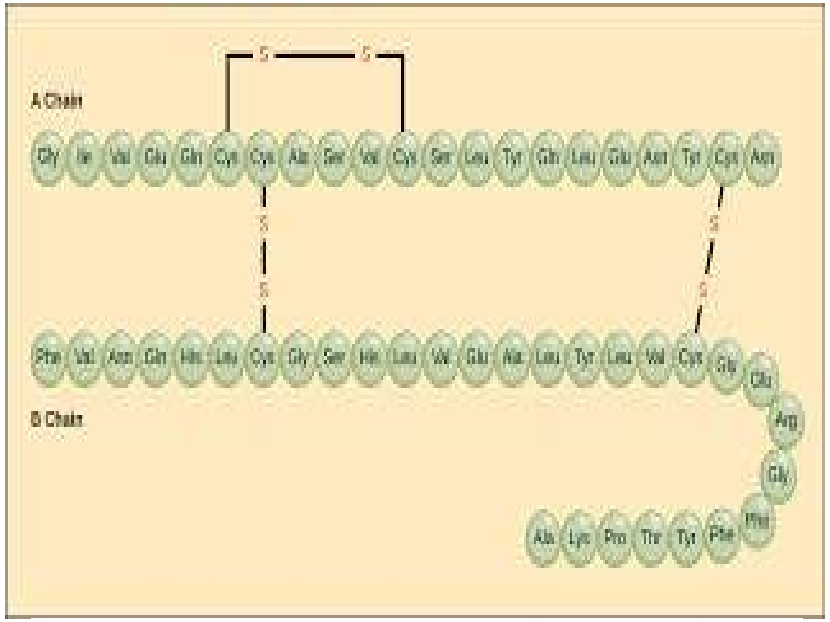
2. Chondroitin sulfates are seen in cartilage and bone.

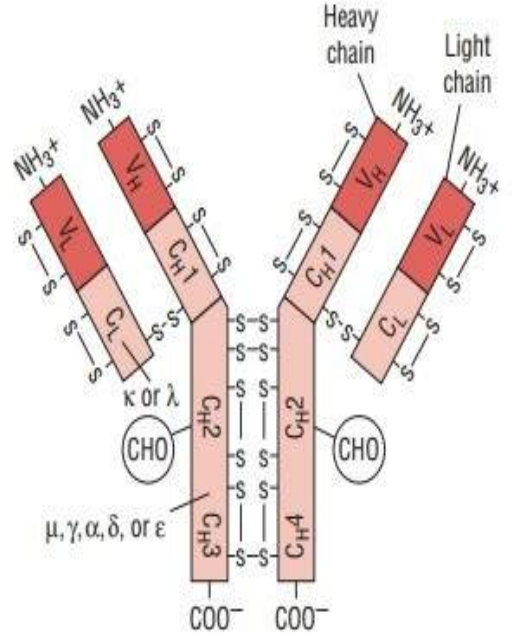
3. Keratin is rich in sulfur, and is present in hair and nails.

4. Many enzymes and peptides contain -SH group at the active site, e.g. glutathione.

5. Co-enzymes derived from thiamine, biotin, pantothenic acid and lipoic acid also contain sulfur.

6. Sulfates are also important in detoxification mechanism. Phenol, skatole, indole and steroids may be detoxicated in the liver with sulphate ions.





THANK YOU